Storytelling as Inverse Inverse Planning

Chandra, Li, Tenenbaum & Ragan-Kelley (2023) — with Borges, *The Garden of Forking Paths*

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Master-Seminar: Theory of Mind

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Scan for notes and slides

Motivation

Two triangles and a circle

🎞 Heider & Simmel (1944)

A 90-second animation of simple shapes moving on a plane.

- viewers see agents with goals
- they judge characters as bully, victim, hero
- no faces, no words — only motion

🤖 Can a computer do this?

First try: simulate rational agents with a planner.

- the agents reach their goals
- but the animation is flat
- no suspense, no surprise, no irony

💡 Question of the paper

What does a storyteller do that a rational agent does not?

Acting *in* vs. acting *out* a situation

Acting *in* a situation

Optimize the character's reward

- agent wants to reach a goal
- best action = shortest path to the goal

Acting *out* a situation

Optimize the audience's belief

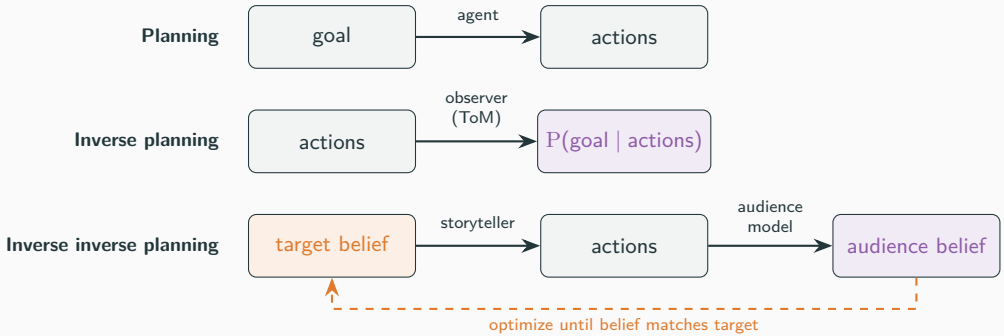
- storyteller wants the audience to *infer* a goal
- best action can be sub-optimal in the world

Stories as "probability design" (Kukkonen, 2014)

A story is a sequence of observations that moves the reader's beliefs over time. Good stories design the journey of those beliefs.

Background

Planning, inverse planning, inverse inverse planning



Key idea

The storyteller plans *through* a model of the audience's theory of mind.

Bayesian inverse planning

Theory of mind as Bayes' rule

$$\mathrm{P}(\text{goal} \mid \text{action}) \propto \mathrm{P}(\text{action} \mid \text{goal})\,\mathrm{P}(\text{goal})$$

- likelihood: a softmax-rational agent
- prior: what goals are plausible

Earlier in this seminar

- 03.12.: Bayesian ToM (Baker et al.)
- 17.12.: LLM-augmented inverse planning
- 14.01.: `memo` hands-on

This paper

Uses inverse planning not to *read* minds, but as the audience model to optimize against.

The audience model: help or hinder? (Ullman et al., 2009)

Grid-world kitchen

- two agents: a strong robot and a weak cheese
- cheese moves succeed only 60% of the time
- robot can push the cheese and a table
- goal tiles: pink and green

Social reward

Robot earns $\rho_{\text{robot}} \cdot r_{\text{cheese}}$ on top of its own reward:

- $\rho > 0$: helping, $\rho < 0$: hindering, $\rho = 0$: indifferent
- two levels of ToM: robot models the cheese

Hypothesis space and observer

$H = \langle G_{\text{cheese}} \in \{p, g\},\ G_{\text{robot}} \in \{p, g, \emptyset\},\ \rho \in \{0, \pm 1, \pm 3\} \rangle, \quad \mathrm{P}(s \to a \mid H) \propto \exp\left(\beta\, Q_c^H(s, a)\right),$
$\mathrm{P}(H \mid s \to a) \propto \mathrm{P}(s \to a \mid H)\, \mathrm{P}(H)$

Inverse inverse planning

Optimizing a script

Script

$\sigma = \big(s_0,\ \sigma_1, \ldots, \sigma_T\big)$,
$\sigma_t = \langle a_t^{\text{robot}}, a_t^{\text{cheese}}, s_t \rangle$

- initial state ("setting") is also optimized
- outcome of each move is chosen too

Objective: depict "helping"

$$f_{\text{help}}(\sigma) = \sum_{t=1}^{T} \mathrm{P}(\rho_{\text{robot}} > 0 \mid \sigma_{1:t})$$

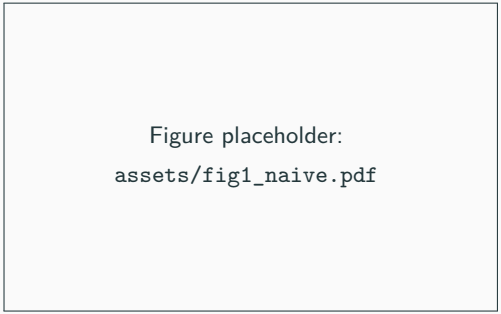
- reward strong belief at every time step
- optimized with beam search

Abstract objective

f_{help} does not fix goals, positions or plot. The optimizer finds a scene that shows helping.

Naïve planning vs. inverse inverse planning

Naïve planning



Cheese already at pink, robot goes to green. Help or indifference? Observer stays unsure.

Inverse inverse planning



Robot pushes cheese to pink, then steps back. Observer is sure: helping.

based on Chandra et al. (2023), Fig. 1 and Fig. 2.

Behavioral experiment 1: do humans agree?

Setup

- 20 seeds $\times$ 3 relations $\times$ 2 methods
- 98 online participants, 15 videos each
- answer: help / hinder / indifferent / unclear

Target	Naïve	Inverse inverse
Helping	6%	**73%**
Hindering	29%	**63%**
Indifferent	75%	73%

Help and hinder: $p < 0.01$. Indifferent: n.s.

Reading

Indifference is easy for a planner (agents ignore each other). Helping is hard: the cheese is often already at its goal.

Miming by "inverse inverse physics"

A hopper pulls a box uphill

- differentiable physics, actor-critic (SHAC)
- audience model: simulate light vs. heavy box, infer weight
- optimize so a light box looks heavy

Figure placeholder:

`assets/fig4a_hopper.pdf`

based on Chandra et al. (2023), Fig. 4.

Behavioral experiment 2 (N = 35)

- same condition pairs: chance (50%)
- honest heavy vs. light: 97% correct
- mime vs. honest light: **95.7%** say mime is heavier
- mime vs. honest *heavy*: **68.6%** say mime is heavier

Storytelling devices as objectives

Plot twist

$$f_{\text{twist}} = \sum_t \begin{cases} \mathrm{P}(\rho > 0 \mid \sigma_{1:t}) & t \le T/2 \\ \mathrm{P}(\rho < 0 \mid \sigma_{1:t}) & t > T/2 \end{cases}$$

Robot "helps" the cheese to pink, then traps it.

Dramatic irony

Audience knows the cheese wants green; robot does not:
$\mathrm{P}(G_c = g) + \mathrm{P}(\rho < 0 \mid G_c = g) + \mathrm{P}(\rho > 0 \mid G_c = p)$

Flashback

$f_{\text{flashback}}(\sigma) = f_{\text{help}}(c(\sigma))$: append a known event, then find a past that recasts it.

Narrative arc + deus ex machina

$$f_{\text{arc}} = \sum_t \sin\left(\tfrac{t}{T} 3\pi\right) \mathbb{E}\left[V^H_{\text{robot}}(s_t)\right] - 0.1\, D_{\mathrm{KL}}(H_{1:t-1} \,\|\, H_{1:t})$$

Fortune falls and rises (Vonnegut's story shapes). KL term keeps goals stable.

The Garden of Forking Paths

Borges (1941): the plot in one slide

The situation

- Yu Tsun, German spy in England, 1916
- knows the town of the new British artillery park
- Captain Madden is hunting him
- no channel to Berlin — the Chief only reads newspapers

What he does

- finds “Stephen Albert” in the phone book
- visits him; they talk about Ts’ui Pen’s labyrinth-novel
- shoots Albert; is arrested and condemned
- the newspapers report the murder
- the town of Albert is bombed

Level 1: the murder *is* inverse inverse planning

Paper	Borges
Storyteller	Yu Tsun
Audience	the Chief in Berlin
Observations	newspaper reports
Target belief	"the target is *Albert*"
Script σ	train, visit, murder, arrest

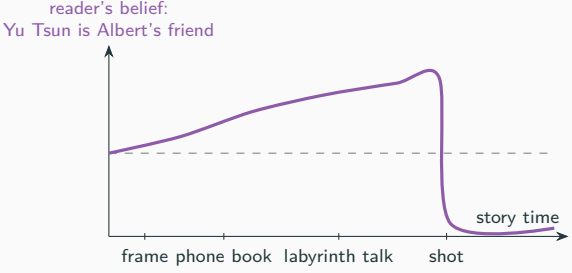
Acting *out*, not *in*

- in the world: very bad action (kills a friend, gets hanged)
- for the belief: the only action that works
- the Chief "solves" it = successful inverse planner

The extreme mime

Like the hopper, the act depicts something it is not: a murder that means a place name.

Level 2: Borges plans the reader's beliefs



illustrative sketch, not data

🔀 Devices from the paper

- **flashback**: Liddell Hart frame — a known delay gets new meaning
- **plot twist**: "I am your friend" then the shot (f_{twist})
- **irony**: Albert says in some futures he is the enemy
- **arc**: terror, hope, calm, murder, triumph

Level 3: the garden is the space of scripts

Ts'ui Pen vs. the optimizer

- Ts'ui Pen's novel takes every branch at once
- the optimizer searches the same tree, keeps one σ
- two battle chapters, two paths, same victory — many σ satisfy one f

The chess riddle

- the novel is about time, but never uses the word
- the reader must infer the concept
- Yu Tsun never sends the word "Albert" either

Show, don't tell

Storytelling = choosing one path through the garden for its effect on what the audience infers.

Discussion

Limits and critique

Stated in the paper

- scales badly: state, hypothesis and script spaces grow fast
- too slow for real-time improvisation
- no emotion in the audience model

Raised by Borges

- **shared hypotheses**: the Chief must already consider "a killing encodes a name"
- **noisy one-shot channel** vs. full noiseless trajectory
- **unreliable narrator**: the editor's footnote contradicts Yu Tsun
- **remorse** is outside any belief objective

Ethics

The same machinery that depicts helping can deceive: legible vs. deceptive motion in robotics.

Discussion questions

1. Write an objective $f(\sigma)$ for Yu Tsun. Which term makes the murder beat every other action?
2. Is the Chief a Bayesian inverse planner? What prior does he need?
3. Can a branching story like Ts'ui Pen's novel be optimized at all — or is it the refusal to choose σ?
4. Is a good plot twist a script that maximizes D_{KL} between beliefs at one time step?
5. Could an LLM serve as the audience model instead of a hand-built inverse planner?



Questions

Sources i

Baker, C. L., Saxe, R., and Tenenbaum, J. B. (2009).
Action understanding as inverse planning.
Cognition, 113(3):329–349.

Borges, J. L. (1941).
The garden of forking paths.
In *El jardín de senderos que se bifurcan*. Sur, Buenos Aires.

Chandra, K., Li, T.-M., Tenenbaum, J. B., and Ragan-Kelley, J. (2023a).
Acting as inverse inverse planning.
In *ACM SIGGRAPH 2023 Conference Proceedings*.

Chandra, K., Li, T.-M., Tenenbaum, J. B., and Ragan-Kelley, J. (2023b).
Storytelling as inverse inverse planning.
In *Proceedings of the Annual Meeting of the Cognitive Science Society*.

Heider, F. and Simmel, M. (1944).
An experimental study of apparent behavior.
The American Journal of Psychology, 57(2):243–259.

Sources ii



Kukkonen, K. (2014).
Bayesian narrative: Probability, plot and the shape of the fictional world.
Anglia, 132(4):720–739.



Ullman, T., Baker, C., Macindoe, O., Evans, O., Goodman, N., and Tenenbaum, J. B. (2009).
Help or hinder: Bayesian models of social goal inference.
In *Advances in Neural Information Processing Systems*, volume 22.

Appendix: implementation details

Grid world

- multi-agent MDP, actions $\{\leftarrow, \rightarrow, \uparrow, \downarrow, \text{stay}\}$
- value iteration, hierarchical softmax policies
- beam search over scripts, $T = 15$ in minutes
- all examples use random seed 0

Physics

- "Luxo lamp" hopper on a hill
- SHAC: policy $\pi(s; \theta)$ and value $V(s; \phi)$
- box weights 0.1 (light) and 0.5 (heavy)
- stumbling: gradient descent on residuals so V dips in $[t_s, t_f]$

Code and videos

`https://people.csail.mit.edu/kach/a2i2p/`